



MOTIVATION

- Safety critical systems:** the system must respect safety constraints at all times.
- Collecting new data is expensive:** robot roll outs are costly and risky to collect.
- Distribution shift is unavoidable:** states seen during testing can be outside the training distribution.
- Overestimating safety is dangerous:** controllers may overestimate the safety of unseen states.
- We propose CCBF:** CCBF is a conservative neural control barrier function that prevents a system from reaching unsafe states and disincentivizes it from reaching out of distribution ones.



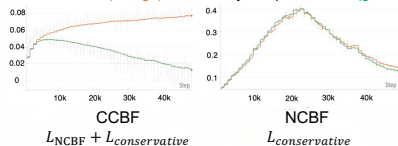
CCBF TRAINING OBJECTIVE

$$L_{CCBF} = L_{NCBF} + L_{conservative}$$

$$L_{conservative} = \frac{w_c}{|X_{safe}|} \sum_{x_i \in X_{safe}} \tau \log \sum_{a_j \in V_x} \exp\left(\frac{B_{\theta}(x_i)}{\tau}\right)$$

B_{θ} – neural CBF with parameters θ
 V_x – set of random actions
 X_{safe} – safe states in the dataset
 w_c – hyperparameter
 τ – temperature

Mean of the barrier values of states reached by taking either dataset actions (orange) or randomly sampled actions (green)



METHOD OVERVIEW

- Offline Dataset:** Safe and unsafe trajectories
- More data is expensive to collect:** collecting unsafe trajectories is costly and risky
- Learn a safety filter:** learn a conservative neural control barrier function to steer the system to safety
- Safe deployment:** filter actions violating safety constraints and leading the system out of distribution of the training data



KEY IDEA

CCBF learns a conservative neural control barrier functions that does not over-estimate the safety of unseen states and actions.



CBF-QP SAFETY FILTER

$$u_{safe}(x) = \arg \min_{u \in U} \|u - u_{ref}(x)\|^2$$

$$\text{s.t. } \nabla B(x)(f(x) + g(x)u) + \alpha(B(x)) \geq 0.$$



The safety filter returns the safe control $u_{safe}(x)$ that is closest to the potentially unsafe reference control $u_{ref}(x)$.



CONSERVATIVE OBJECTIVE

- Keep the system from reaching the failure set.
- Discourage actions unseen in the dataset.

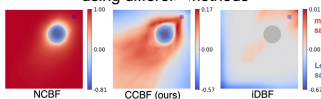


EXPERIMENTAL RESULTS

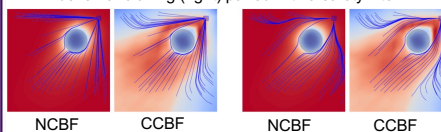
Dataset trajectories



Control barrier functions learned using different methods

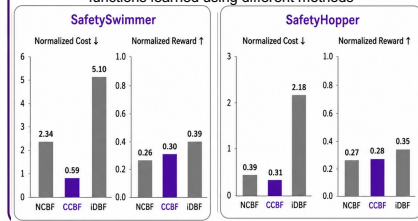


Trajectories generated using PID controller (left) and behavior cloning (right) paired with a safety filter



EXPERIMENTAL RESULTS

Reward and cost averaged over 5 nominal controllers learned using safe offline RL and paired with neural control barrier functions learned using different methods



TAKEAWAY

CCBF outperforms baselines, keeping the system safe and encouraging it to stay within the support of its training data.